Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	An AI-Based Digital Twin Framework for Energy Optimization in Energy-Intensive Manufacturing Processes and Facilities			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), **AI (√)**, Next-gen Communication ( ), **Advanced Robotics & Manufacturing (√)**, Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	*Production Process Modeling / Simulation*	*Software*	*Software Solutions*
Concept of Proposed Technology	Energy optimization is an increasingly critical challenge for energy-intensive manufacturing processes and facilities from economic, technical, and policy perspectives. Energy costs often account for a substantial share (can be 20% or more) of total production costs [1], while recent geopolitical and market uncertainties have further increased price volatility. Technically, advanced manufacturing processes are highly sensitive to operating conditions; even minor disturbances can affect equipment reliability and product quality. This requires moving beyond simple energy reduction toward identifying optimal operating conditions that balance energy use, productivity, and quality. At the same time, global initiatives such as the European Union's Carbon Border Adjustment Mechanism (CBAM), RE100, and Digital Product Passports (DPP) are accelerating the shift from conventional energy management toward carbon- and life-cycle-oriented optimization. However, current practices largely rely on equipment-level monitoring, post-hoc analysis, and operator experience, which limit process-level understanding and predictive decision-making. The integration of AI and digital twins offers a pathway from passive monitoring to predictive simulation and optimization, enabling process- and product-level energy optimization and structured life-cycle data			

	management. To effectively support optimization and decision-making, a digital twin must be capable of explaining and predicting how target systems respond to diverse operating conditions and operational interventions. In real-world manufacturing environments, however, collecting high-quality data that sufficiently covers a wide range of operating regimes and intervention scenarios is often impractical. Under such conditions, conventional data-driven machine learning models are vulnerable to extrapolation errors and context shifts, limiting their reliability and industrial applicability [2]. To address these challenges, this project proposes the development of an AI-based digital twin framework for energy-intensive manufacturing processes and facilities. Unlike conventional digital twins that focus primarily on visualization and monitoring, the proposed framework is designed to generate actionable operational intelligence, including energy peak prediction, prevention of equipment shutdowns, and identification of critical power thresholds affecting product quality. The framework integrates physics-based system modeling with physics-informed and causation-based machine learning, uncertainty quantification, and probabilistic active learning to ensure robust and explainable prediction under data-limited and dynamically changing conditions. Building on this foundation, reinforcement learning is employed to derive optimal process and energy operation strategies within the digital twin, while digital-twin-based anomaly detection enables early identification of abnormal energy usage and power-quality events that may lead to safety incidents or production losses. The proposed framework is modular, computationally efficient, and extensible, supporting a wide range of applications including energy and process optimization, anomaly detection, safety management, and predictive maintenance.
Rationale for Joint R&D Project	The proposer has extensive experience in developing and applying advanced methodologies, such as physics-informed machine learning, causal learning, uncertainty-aware analytics, and digital twins, to building and facility design, operation, and energy management. These approaches have been successfully validated in real-world and simulation environments, demonstrating their effectiveness in improving energy efficiency and operational decision-making [3-8]. Extending these methodologies to energy-intensive manufacturing processes is expected to generate significant additional impact, given the higher complexity, tighter operational constraints, and greater economic stakes involved. The successful development and deployment of the proposed digital twin framework critically depends on access to real operational data and practical knowledge of manufacturing environments. Through collaboration with an industry partner possessing extensive smart factory experience and real-world process data, this project creates a strong synergy between theory-driven innovation and field-proven industrial practice. Building on this foundation, the proposed joint R&D will advance current capabilities from retrospective analysis and short-term cost prediction toward future-oriented scenario analysis and optimal condition discovery. Through this collaboration, the project aims to develop internationally compatible energy and carbon data models, as well as AI and digital twin frameworks applicable to global manufacturing sites, thereby maximizing both technological and industrial impact.
Key Words	1. AI-Based Digital Twin 2. Physics-Informed and Causation-Based Machine Learning 3. Uncertainty Quantification and Active Learning 4. Reinforcement Learning 5. Anomaly Detection

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

Goal: Development of an AI-Based Digital Twin Framework for Energy Optimization in Energy-Intensive Manufacturing Processes and Facilities

Objective 1. Development of an AI-Based Digital Twin Framework (O1) – Establishing a robust and extensible digital twin framework capable of integrating physical models and AI for energy-intensive systems and processes.

Objective 2. Development of Analysis and Optimization Tools (O2) – Enabling predictive analysis, decision support, and energy optimization based on developed digital twins.

Objective 3. Testing and Evaluation (O3) – Validating the proposed framework and tools using real-world data and testbed environments.

R&D Target Specifications

Quantitative Performance Specifications

To ensure the practical effectiveness and industrial relevance of the proposed AI-based digital twin framework, clear qualitative and quantitative performance specifications are defined. The proposed approach integrates physics-informed machine learning, causal learning, uncertainty quantification, and active learning to achieve robust and explainable predictive performance in real-world manufacturing environments characterized by changing operating conditions and limited data availability. Moving beyond monitoring-oriented digital twins, the framework is designed to generate actionable decision support, including proactive energy peak management, advance identification of power thresholds critical to product quality, and early detection of power-quality anomalies that may lead to equipment shutdowns or safety incidents. Based on these capabilities, quantitative performance targets are defined to objectively evaluate improvements in prediction accuracy, robustness under distribution shifts, data efficiency, cost reduction through optimization, and anomaly detection performance, as summarized in the following table.

Specifications	Evaluation	Target	Metrics
Digital Twin In-Distribution (ID) Prediction Performance[1]	Comparison against digital twins generated with conventional white-box approaches	15% improvement	RMSE, EMD
Digital Twin Out-of-Distribution (ODD) Prediction Performance[1]	Comparison against digital twins generated with conventional black-box approaches	20% improvement	RMSE, EMD
Quality of Data Collected through Active Learning	Comparison against passive data collection	20% improvement	JSD
Cost Saving through Optimization	Comparison against schedule-based approaches	3% cost saving	Cost (Korean Won)
Anomaly Detection Rate	Comparison against conventional threshold-based approaches	5% improvement	F1-score

[1] Prediction performance refers to energy consumption, power peaks, and key process variables.

Qualitative Performance Specifications

The proposed AI-based digital twin framework is designed to be highly generalizable and flexible, enabling its application across a wide range of energy-intensive manufacturing processes and facilities. Its modular architecture supports rapid re-configuration to accommodate different systems,

	operating conditions, and levels of data availability. By leveraging physics-informed machine learning, causal learning, uncertainty quantification, and active learning, the framework demonstrates robust and stable predictive performance even in real-world industrial environments characterized by missing data and frequent changes in operating conditions. To support reliable and trustworthy decision-making, the framework explicitly quantifies predictive uncertainty and provides explainable insights grounded in physical and causal process relationships. Moving beyond passive monitoring, the digital twin enables the generation of actionable operational guidance, including proactive energy peak management and advance identification of critical power thresholds required to maintain stable product quality. In addition, the framework supports near real-time computation, making it suitable for deployment in operational settings. It is compatible with a wide range of applications, including anomaly detection, process optimization, safety validation, and predictive maintenance, and is inherently expandable. In particular, the framework enables early detection of power quality anomalies in high-power equipment, establishing a practical safety management standard to prevent major incidents such as equipment shutdowns or fire hazards.
R&D Plan	**Overall Research Approach** The proposed R&D adopts a system-oriented and data-informed research approach to develop, validate, and demonstrate an AI-based digital twin framework for energy optimization in energy-intensive manufacturing processes and facilities. The research is structured around three tightly coupled objectives: (i) development of a robust and extensible digital twin framework integrating physics-based and data-driven intelligence; (ii) development of advanced analysis and optimization tools to support predictive and decision-oriented energy management; and (iii) validation of the proposed framework using real-world industrial data and testbed environments. A key principle of the proposed approach is to move beyond conventional monitoring-oriented digital twins toward an intelligent, decision-support-oriented digital twin capable of robust prediction, uncertainty-aware optimization, and proactive operational guidance. To achieve this, the research integrates physics-based system modeling, physics-informed and causation-based machine learning, uncertainty quantification, and active learning within a unified framework. The R&D plan is designed to ensure scalability, explainability, and real-time applicability, while explicitly addressing challenges related to limited data availability, changing operating conditions, and safety-critical decision-making. **Objective 1: Development of a Digital Twin Framework** **T1.1. Development of a Physics-Based Co-Simulation Environment** A physics-based co-simulation environment will be developed to represent energy-intensive manufacturing systems across multiple physical domains (e.g., electrical, thermal, mechanical). This environment will enable the integration of subsystem models operating at different temporal and spatial scales. Standardized interfaces will be designed to allow seamless coupling between simulation engines, real-time data streams, and AI components. The co-simulation environment will serve as the backbone of the digital twin, enabling synchronized simulation, analysis, and validation. **Methodology:** • Define system boundaries and interfaces • Design co-simulation architecture for multi-domain integration • Validate numerical stability and computational efficiency **Milestone:** Operational co-simulation environment. **T1.2. Development of System and Process Models Using Modelica** System- and process-level models will be developed using Modelica to represent key equipment, processes, and energy flows within target manufacturing facilities. These models will incorporate first-principles physics to capture causal relationships between operating conditions, energy consumption, and process outcomes. Modular model structures will be adopted to support reuse and reconfiguration across different applications. **Methodology:**

- Develop parameterized Modelica models for high-power equipment and processes
- Calibrate models using available operational data
- Validate models under nominal operating conditions

Milestone: Library of validated Modelica-based system and process models.

T1.3. Development of a Physics-Informed and Causation-Based ML Pipeline

To overcome data limitations and improve robustness under changing conditions, a machine learning pipeline integrating physics-informed constraints and causal structure learning will be developed. This pipeline will combine simulation data from the physics-based models with real-world operational data to learn predictive models that generalize beyond observed regimes.

Methodology:

- Embed physical hard/soft constraints into ML architectures
- Learn causal representations of system behavior
- Evaluate robustness under extrapolation and context shifts

Milestone: Physics-informed and causation-based ML pipeline.

T1.4. Development of an Uncertainty Quantification Pipeline

An uncertainty quantification (UQ) pipeline will be developed to quantify predictive uncertainty arising from data sparsity, model mismatch, and operational variability. This capability is critical for reliable decision-making and risk-aware optimization.

Methodology:

- Develop probabilistic deep learning models (e.g., Bayesian or ensemble-based methods)
- Quantify predictive confidence intervals
- Integrate uncertainty outputs into decision modules

Milestone: Operational UQ pipeline producing uncertainty-aware predictions.

Objective 2: Development of Analysis and Optimization Tools

T2.1. Development of an Active Learning Pipeline

A probabilistic active learning pipeline will be developed to guide the acquisition of additional high-value data needed to improve digital twin performance. This approach will identify operating conditions or interventions that maximize information gain while minimizing operational risk.

Methodology:

- Define information-theoretic acquisition functions
- Incorporate uncertainty and operational constraints
- Evaluate data efficiency improvements

Milestone: Active learning pipeline integrated with digital twin.

T2.2. Development of a Reinforcement Learning Pipeline for Process Optimization

A reinforcement learning (RL) pipeline will be developed to derive optimal control and operational strategies for energy optimization while maintaining product quality and system safety. Training will be conducted primarily within the digital twin to avoid risks associated with real-world exploration.

Methodology:

- Define state, action, and reward structures
- Train RL agents using the digital twin
- Evaluate performance against baseline strategies

Milestone: RL-based optimization module.

T2.3. Development of a Digital-Twin-Based Anomaly Detection Methodology

A digital-twin-based anomaly detection methodology will be developed to identify abnormal energy usage patterns and power-quality issues that may lead to equipment shutdowns or safety incidents.

Methodology:

- Model normal operating behavior using the digital twin

- Detect deviations using residual and probabilistic methods
- Validate detection performance using historical and synthetic anomalies

Milestone: Anomaly detection module capable of early fault identification.

Objective 3: Testing and Evaluation

T3.1. Preprocessing and Analysis of Existing Industrial Data

Operational data provided by the SME partner will be preprocessed, cleaned, and analyzed to identify key patterns, data gaps, and modeling requirements.

Milestone: Curated dataset ready for model development.

T3.2. Development of a Site-Specific Digital Twin

A site-specific digital twin will be constructed using the available data and developed models, representing a real manufacturing facility.

Milestone: Operational digital twin of the target site.

T3.3–T3.5. Integrated Testing and Validation

The ML pipelines, optimization tools, and anomaly detection methods will be systematically tested within the digital twin and validated using a physical testbed environment. Performance will be evaluated against predefined quantitative and qualitative specifications.

Milestones:

- Verified predictive performance
- Demonstrated energy cost reduction
- Validated safety and anomaly detection capabilities

Timeline and Deliverables

The project will be executed in phased stages over the project period, progressing from framework development (Year 1), to advanced analytics and optimization (Year 2), and finally to integrated validation and demonstration (Year 3). Key deliverables include the digital twin framework, AI pipelines, validated optimization tools, and demonstration results from real-world and testbed environments.

Objectives & Tasks	Year 1				Year 2				Year 3			
O1. Development of a Digital Twin Framework												
T1.1. Develop a Physics-Based Co-Simulation Environment	■	■	■	■								
T1.2. Develop System/Process Models Using Modelica			■	■	■	■						
T1.3. Develop a Pipeline for Physics-Informed and Causation-Based ML	■	■	■	■	■	■	■	■				
T1.4. Develop a Pipeline for Uncertainty Quantification		■	■	■	■	■	■	■	■			
O2. Development of Analysis and Optimization Tools												
T2.1. Develop an Active Learning Pipeline for the Framework						■	■	■	■			
T2.2. Develop a Reinforcement Learning Pipeline for Process Optimization						■	■	■	■			
T2.3. Develop a Digital-Twin-Based Anomaly Detection Methodology						■	■	■	■			
O3. Test and Evaluation												
T3.1. Preprocess and Analyze Existing Data from the SME partner	■	■	■	■								
T3.2. Develop a Digital Twin of a testbed with the Existing Data			■	■	■	■	■	■				
T3.3. Test and Evaluate the Pipelines from T1.3 & 1.4 in the Digital Twin							■	■	■	■		
T3.4. Test and Evaluate the Pipelines from T2.1 – 2.3 in the Digital Twin								■	■	■	■	
T3.5. Test and Evaluate the Digital Twin with the testbed										■	■	■

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will take the lead in the scientific and methodological development of the proposed AI-based digital twin framework. Specifically, the research institute will be responsible for all research and development activities under **Objective 1** (Development of a Digital Twin Framework) and **Objective 2** (Development of Analysis and Optimization Tools). Under **Objective 1**, the research institute will design and implement the core digital twin architecture, including the physics-based co-simulation environment, system- and process-level models using Modelica, and advanced machine learning pipelines. This includes the development of physics-informed and causation-based machine learning methodologies to ensure robust prediction under limited data availability and changing operating conditions, as well as uncertainty quantification techniques to support reliable and risk-aware decision-making. Under **Objective 2**, the research institute will develop advanced analytics and optimization tools, including probabilistic active learning for data-efficient model improvement, reinforcement learning-based process optimization strategies, and digital-twin-based anomaly detection methodologies. These tools will be designed to operate within the digital twin environment and to generate actionable operational insights while ensuring explainability, robustness, and computational efficiency. In addition, the research institute will jointly participate in **Objective 3** (Testing and Evaluation) with the industry partner, contributing to data analysis, model calibration, performance evaluation, and interpretation of validation results. The research institute will also be responsible for ensuring scientific rigor, documenting methodologies, defining performance metrics, and disseminating research outcomes through technical reports and academic publications. **Summary** • Development of a Physics-Based Co-Simulation Environment (Lead T1.1) • Development of System/Process Models Using Modelica (Lead T1.2) • Development of AI/ML Methodologies and Pipelines (Lead T1.3, T1.4, T2.1, T2.2, T2.3) • Development of a Digital Twin of a test bed (Co-Lead T3.2; Support T3.1) • Test and Evaluate AI/ML Methodologies and Pipelines (Co-Lead T3.3, T3.4; Support T3.5)
Expected R&R of Korean SME Partner	The industry partner will play a critical role in ensuring the practical relevance, industrial applicability, and commercialization potential of the proposed technology. While jointly leading **Objective 3** (Testing and Evaluation), the industry partner will primarily lead industrial data integration, system implementation, field validation, and solution commercialization activities. **Technical Contributions** The industry partner will provide key technical capabilities for integrating real-world factory environments with the proposed digital twin framework. This includes the development of industrial-standard-based data integration technologies to collect and unify energy and process data using interoperable protocols such as OPC UA and Asset Administration Shell (AAS). The partner will also contribute domain knowledge for analyzing energy consumption patterns in relation to production outcomes, enabling the identification of abnormal energy usage and inefficient process patterns. In addition, the industry partner will provide a way to automatically extract system/process-related physical properties from CAD and relevant materials to support digital twin model generation. By extracting relevant system, equipment, and layout information directly from high-power equipment specifications and factory drawings, this will significantly improve the speed, scalability, and accuracy of digital twin modeling. **Development Tasks** The industry partner will provide process-line and high-power equipment design data required for

	constructing initial digital twin models and will support the configuration of site-specific digital twins. Collected factory real-world data will be cleaned, standardized, and transformed into formats suitable for AI model training and validation. The partner will also develop user-facing components, including UI/UX for actionable insights, translating AI model predictions into practical operational guidance such as energy peak response actions, equipment shutdown prevention, and quality assurance alerts. **Testing and Validation Activities** The industry partner will lead demonstration and validation activities using real testbeds. This includes operating testbeds, validating key performance indicators (KPIs), and evaluating the consistency between real-time factory data and digital twin predictions. The partner will also conduct user acceptance testing with on-site operators to assess the usability, effectiveness, and practicality of the generated actionable insights. **Resource Commitment** The industry partner will commit existing and newly acquired datasets from energy-intensive industrial environments, along with professional personnel required for system integration, data engineering, and field operation. Where applicable, the partner will provide modeling support tools, APIs, and software components to facilitate efficient digital twin deployment and system integration. **Collaboration and Commercialization Activities** Beyond the R&D phase, the industry partner will collaborate with the research institute to define business models for post-project commercialization, including subscription-based digital twin solutions targeting energy-intensive manufacturing sectors. Joint efforts will also include promoting project outcomes through exhibitions, conferences, and industry networks, as well as identifying and engaging new industrial customers for large-scale deployment.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical and Intellectual Outcomes** This project is expected to deliver core AI and digital twin technologies that fundamentally advance the state of intelligent energy and process management in energy-intensive manufacturing environments. From an intellectual perspective, the project will generate original methodological contributions by integrating physics-informed machine learning, causal learning, uncertainty quantification, and active learning into a unified digital twin framework. These contributions address a critical limitation of conventional AI approaches, namely, their vulnerability under limited data, missing values, and changing operating conditions, and establish robust, explainable, and uncertainty-aware predictive models applicable to extreme manufacturing environments. Beyond monitoring, through reinforcement learning and twin-based anomaly detection, the framework enables intelligent decision-making that simultaneously optimizes energy cost, operational stability, and product quality through physics- and data-consistent reasoning. **Commercialization Potential and Business Model** The proposed technology is designed with commercialization as a core objective. The primary commercialization pathway is the expansion into a subscription-based SaaS digital twin platform targeting energy-intensive industries with high electricity cost exposure, such as steelmaking, chemical processing, and data centers. The platform will provide modular services including energy and process optimization, anomaly detection, predictive maintenance, and automated energy and carbon reporting. By ensuring compatibility with existing industrial infrastructure, such as power meters, sensors, PLCs, and legacy systems, the proposed solution lowers adoption barriers for small and mid-sized manufacturers. Leveraging the industry partner's existing customer network and deployment experience, the project outcomes can be rapidly transitioned into market-ready solutions. This

	positions the technology not only as a research output but as a revenue-generating, scalable product aligned with post-project sustainability. **Industrial and Economic Impact** The project is expected to deliver tangible economic value by reducing energy costs, mitigating operational risks, and improving productivity in national core manufacturing industries. Proactive management of energy peaks and power-quality anomalies can prevent equipment shutdowns and safety incidents, whose economic impact, such as production losses per hour of downtime, far exceeds direct energy savings. By translating shutdown and accident prevention into quantifiable cost avoidance, the technology demonstrates clear return on investment and adoption necessity. At the industrial scale, the project contributes to enhanced cost competitiveness and export capacity by reducing energy peak charges, minimizing quality defects caused by voltage disturbances, and stabilizing production. Moreover, precise energy optimization directly supports ESG management and national greenhouse gas reduction targets by eliminating unnecessary power consumption and enabling carbon-aware operations. **Qualitative and System-Level Impact** Beyond measurable performance gains, the project is expected to drive a structural transformation in manufacturing operations. The proposed framework enables a shift from equipment-level, retrospective energy monitoring toward an integrated, predictive, and optimization-driven operational paradigm that jointly considers energy, production, process conditions, and carbon emissions. This transition reduces operational uncertainty and enables stable, simulation-based decision-making rather than reactive control. Furthermore, the project delivers a scalable and extensible framework that is not limited to a single factory or process. By establishing a generic digital twin and AI foundation, the outcomes can evolve into industry-specific foundation models aligned with government strategies for manufacturing digital transformation. This democratizes access to advanced energy and carbon management capabilities for small and mid-sized manufacturers, narrowing technological gaps and promoting inclusive industrial growth.

6. References

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